

Numerical Analysis

Lab 9

Giuseppe Calì

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Trigonometric interpolation and Least Square Method

This lab deals with trigonometric and least square method.

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Trigonometric interpolation I

Consider the case of a periodic function $f(x)$ with period 2π and with $f(0) = f(2\pi)$. The $n + 1$ values at equally spaced nodes

$$x_j = 2\pi j / (n + 1) \quad j = 0, \dots, n$$

are known.

The constant step size is

$$h = 2\pi / (n + 1)$$

.

The Lagrange interpolator may not approximate well far from the nodes, as it is a polynomial of degree n and generally not periodic.

Goal: Construct a periodic interpolator $\tilde{f}(x)$.

Trigonometric interpolation II

The trigonometric interpolant $\tilde{f}$ is obtained by a linear combination of sines and cosines. Setting $M = \frac{n}{2}$ with n even:

$$\tilde{f}(x) = \frac{a_0}{2} + \sum_{k=1}^M (a_k \cos(kx) + b_k \sin(kx))$$

with complex unknown coefficients a_k for $k = 0, \dots, M$ and b_k for $k = 1, \dots, M$. $\tilde{f}(x)$ is a 2π -periodic function characterized by $M + 1$ frequencies $k = 0, \dots, M$ and coefficients a_k and b_k .

Trigonometric interpolation III

Using the Euler notation, the trigonometric interpolant can be written as:

$$\tilde{f}(x) = \sum_{k=-M}^M c_k e^{ikx}$$

with $c_k = \frac{1}{2}(a_k - ib_k)$ and $c_{-k} = \frac{1}{2}(a_k + ib_k)$ for $k = 0, \dots, M$.

The coefficients c_k are complex and the interpolant is a linear combination of complex exponentials.

The interpolation conditions are:

$$\tilde{f}(x_j) = \sum_{k=-M}^M c_k e^{ikjh} = f(x_j) \quad \text{for } j = 0, \dots, n$$

which gives $n + 1$ equations for the $2M + 1$ unknowns c_k .

Trigonometric interpolation IV

With some algebra, the coefficients c_k can be expressed in terms of the $f(x_j)$ (obtaining the **DFT**, Discrete Fourier Transform):

$$c_k = \frac{1}{n+1} \sum_{j=0}^n f(x_j) e^{-ikjh} \quad \text{for } k = -M, \dots, M$$

or in matrix form:

$$\begin{pmatrix} c_{-M} \\ \vdots \\ c_M \end{pmatrix} = \frac{1}{n+1} \begin{pmatrix} e^{iMh0} & \dots & e^{iMhn} \\ \vdots & \ddots & \vdots \\ e^{-iMh0} & \dots & e^{-iMhn} \end{pmatrix} \begin{pmatrix} f(x_0) \\ \vdots \\ f(x_n) \end{pmatrix}$$

$$\mathbf{c} = \mathbb{T} \mathbf{f}$$

where $\mathbf{c} \in \mathcal{R}^{n+1}$, $\mathbf{f} \in \mathcal{R}^{n+1}$ and $\mathbb{T}$ is the Fourier matrix with coefficients:

$$T_{kj} = \frac{1}{n+1} e^{-ikjh} \quad \text{for } k = -M, \dots, M \quad \text{and } j = 0, \dots, n$$

Trigonometric interpolation V

The Inverse Discrete Fourier Transform (IDFT) is:

$$f(x_j) = \sum_{k=-M}^M c_k e^{ikjh} \quad \text{for } j = 0, \dots, n$$

or in matrix form:

$$\mathbf{f} = \mathbb{T}^{-1} \mathbf{c}$$

where $\mathbb{T}^{-1}$ is

$$(\mathbb{T}^{-1})_{kj} = e^{ikjh} \quad \text{for } k = -M, \dots, M \quad \text{and } j = 0, \dots, n$$

Theorem 9.1 (Interpolation error)

If f is 2π -periodic and admits continuous and 2π -periodic derivative up to order s included, then the error of the trigonometric interpolant $\tilde{f}$ is bounded by:

$$\max_{x \in [0, 2\pi]} |f(x) - \tilde{f}(x)| \leq C \frac{1}{n^s}$$

for some constant C . If this holds $\forall s > 0$, then $\tilde{f}$ converges to f exponentially.

Exercise 9.1

- a Implement a function that computes the coefficient of the Fourier trigonometric interpolant.
- b Considering $M = 25$ and evenly distributed nodes in the interval $[0, 2\pi]$, find the trigonometric interpolant of the following functions and study the convergence order of the interpolation motivating the results:

$$f(x) = 1 - \frac{4x^2}{\pi^2} + \frac{4x^3}{\pi^3} - \frac{x^4}{\pi^4}$$

$$f(x) = \text{interp1}([0 \ 2\pi/3 \ 4\pi/3 \ 2\pi], [0 \ 1 \ -1 \ 0], x, 'linear')$$

$$f(x) = 1.0(x > \pi/2 \text{ and } x < 3 * \pi/2)$$

$$f(x) = \sin(x + \cos(5x))$$

Exercise 9.2

Repeat point b of the previous exercise using MATLAB command `interpft`.

Exercise 9.3 (Exam exercise)

Consider the function

$$f(x) = \cos(5x + \sin(5x))$$

- a Use the provided function `int_trigonometrica.m` to compute the Fourier coefficients of the trigonometric interpolant.
- b Study the convergence of the trigonometric interpolant to the function $f(x)$. Motivate the results.
- c Compare the trigonometric interpolant with the Lagrange interpolant of degree 10 and 15. Motivate the results.

Lagrangian Expression of the Trigonometric Interpolant I

In the Lagrangian polynomial the interpolant was expressed as a basis function times coefficients that were the function values at the nodes: We can express the trigonometric interpolant in a similar way:

$$\tilde{f}(x) = \sum_{j=0}^n f(x_j) \varphi_j^T(x)$$

with the Lagrangian basis functions:

$$\varphi_j^T(x) = \frac{1}{n+1} \sum_{k=-M}^M e^{ik(x-jh)} \quad \text{for } j = 0, \dots, n$$

Exercise 9.4

Given $n = 4$ and $h = 2\pi/(n + 1)$ find the trigonometric Lagrangian basis functions $\varphi_j^T(x)$ for $j = 0, \dots, 4$ and compare it with the Lagrangian basis functions seen in the previous laboratory.

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Least Square Method I

The least square method is used to approximate a function $f(x)$ by a polynomial of degree $m \ll n$, with n the number of nodes. The polynomial is obtained by minimizing the error function in the least square sense:

$$\min_{p \in \mathcal{P}_m} \sum_{i=0}^n (y_j - p(x_j))^2$$

where $\mathcal{P}_m$ is the space of polynomials of degree m . The general form of the polynomial is:

$$p(x) = \sum_{k=0}^m a_k x^k$$

If $m = 1$, the polynomial is called regression line and the error is minimized by the line that passes through the points $(x_j, f(x_j))$. We need to find the coefficients a_0 and a_1 .

The problem can be stated as follows:

$$\Phi(b_0, b_1) = \sum_{j=0}^n (y_j - (b_0 + b_1 x_j))^2$$

Find (a_0, a_1) such that

$$\Phi(a_0, a_1) = \min_{b_0, b_1} \Phi(b_0, b_1)$$

Least Square Method III

The minimum is found by setting the derivatives of Φ with respect to b_0 and b_1 to zero:

$$\frac{\partial \Phi}{\partial b_0} = 0 \quad \text{and} \quad \frac{\partial \Phi}{\partial b_1} = 0$$

which given the solution:

$$a_0 = \frac{\sum_{j=0}^n y_j \sum_{j=0}^n x_j^2 - \sum_{j=0}^n x_j \sum_{j=0}^n y_j x_j}{D}$$

$$a_1 = \frac{(n+1) \sum_{j=0}^n x_j y_j - \sum_{j=0}^n x_j \sum_{j=0}^n y_j}{D}$$

with $D = (n+1) \sum_{j=0}^n x_j^2 - (\sum_{j=0}^n x_j)^2$.

Exercise 9.5

Generate a random sample of $n + 1$ points (x_j, y_j) .

- a Compute the coefficients of the regression line. Plot the points and the regression line.
- b Repeat the exercise using the MATLAB command `polyfit`.
- c Compute the mean square error of the regression line. Verify that the error is minimized.

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3 Homework

Homework I

References I